

Converting a PDF Textbook to be Accessible

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- 1978** T_EX
- 1990** Americans with Disabilities Act (ADA).
Title II: Government entities may not discriminate against those with disabilities.
- 1993** PDF
- 2016** University of North Dakota adopts A^P_EX Calculus.
- 2017–2020** PDF 2.0 (better math support).
- 2019** L^AT_EX team begins improving accessibility in earnest
- 2024-04-26** US Dept of Justice: ADA Title II means that public entities must make documents accessible.
Most entities get two years. We (DoJ) get three years.
- 2026-04-20** Everyone gets an extra year.

Preamble

Images

Tables

All else

Topic	Time spent (%)			How accessible (%)
	on the textbook	today	on your own	
Preamble	3	10	<1	85
Images	20	30	25	5
Tables	17	30	25	5
All else	60	30	50	5
	100	100	100	100

Update $\text{\LaTeX}$

```
\DocumentMetadata{
  lang=en,
  tagging=on,
  pdfstandard=ua-2,
% check-tagging-status,
  tagging-setup={
%   math/alt/use,
    math/setup=mathml-SE
  }
}
\RequirePackage{unicode-math}
```

compile with Lua $\text{\LaTeX}$, and most of the work is done
texdoc documentmetadata-support-code

Use $\text{\LaTeX}$ so that the output looks how you want it
and the input reflects the meaning.
(Don't use `tabular` to position boxes.)

Make use of *semantic* macros and environments:
`\textbf{Example 3.1.4 Finding Extreme Values}`
should be
`\begin{example}{Finding Extreme Values}`

- `\includegraphics[alt={...}]`
- `\tikz[alt={...}]`
- `\begin{tikzpicture}[alt={...}]`
- `\begin{picture}[alt={...}]`

Keep it brief and objective; exposition won't be seen by most students.

Alternatives to `alt={...}` are

- `artifact` (image is decorative only) 
- `actualtext={...}` (image is text in disguise)



- Usually not seen
- Specific, accurate, objective, neutral
- What does my audience need from this picture?
- Do not begin with “image”
- Do not repeat the caption
- At most 120 characters or 1–2 sentences.
If not possible, have the alt text point to a separately written description
- `\setkeys{Gin}{alt={...}}`
- “Formulas need alt text” $\leftrightarrow$ option `math/alt/use`
- Have someone else write the alt text?

- `\tagpdfsetup{table/header-rows={...},table/header-columns={...}}`
outside of the tabular
(lasts until the group `\ends`)
- `\tagpdfsetup{table/multirow={number of rows}}`
- Interface will probably change
- `texdoc latex-lab-table`

- `enumitem` & `enumerate` → `enumext` or `templates` (`texdoc blocks-doc`)
- `margin[fit|fix|note]` → `margin[alia|par]`
- `titlesec` →
 - `\@startsection` (`texdoc latex2e`, §6.8)
 - `templates` (`texdoc latex-lab-sec-template`)
- Beamer → `ltx-talk` (tomorrow)
- Some packages need some (or a lot of) help (`tcolorbox`, `qrcode`, `media9`, ...) use option `check-tagging-status` or <https://latex3.github.io/tagging-project/tagging-status/>

To verify that a PDF is accessible:

Screen reader: (NVDA or JAWS) and (Adobe, FoxIt, or Firefox)

VeraPDF: <https://verapdf.org/software>



has command line version; L^AT_EX team



no GUI for PDF errors

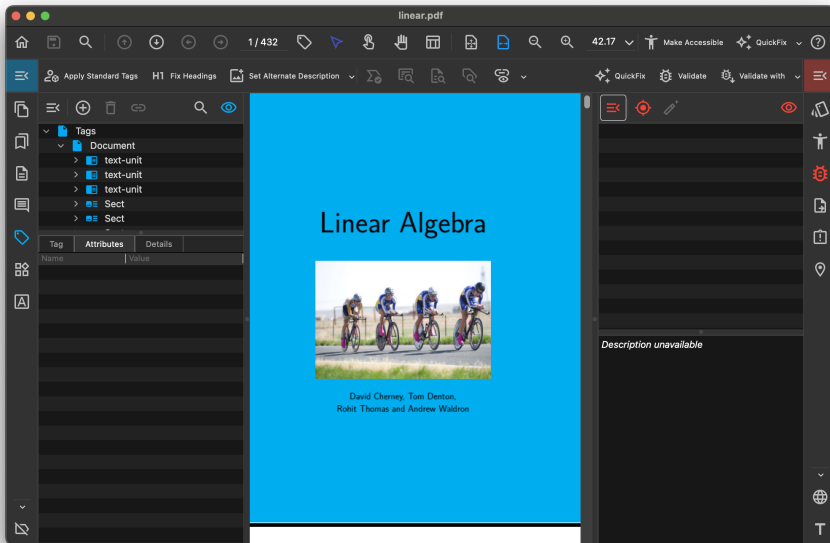
PDFix: <https://pdfix.net>

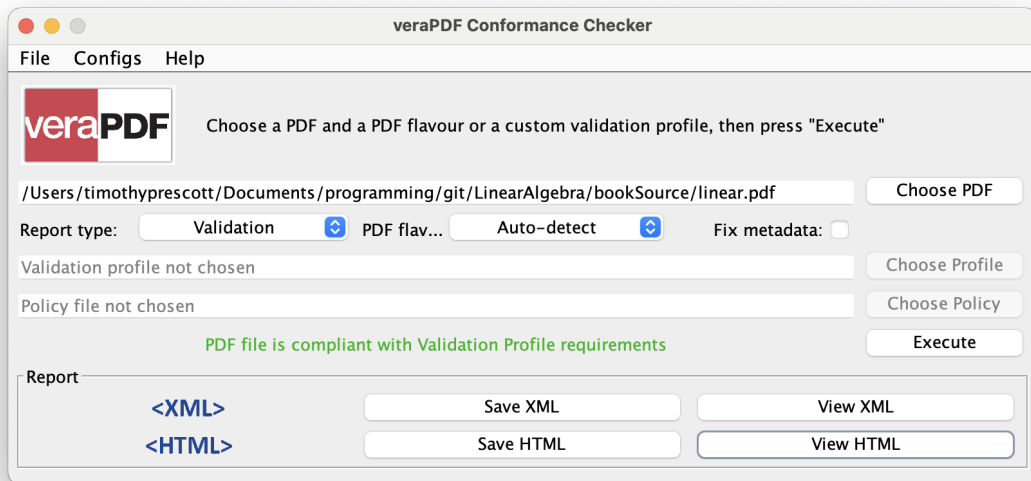


GUI



no command line





- <https://latex3.github.io/tagging-project/documentation/prototype-usage-instructions.html>
- Compiling needs more:

Stat	Factor
Time	6.7
RAM	4.4
Output size	1.3
strings	11
hash	11

Try `\DocumentMetadata{tagging=off}` for draft runs

Math	before	after
$a^2 + b^2 = c^2$	ay two plus bee two equals cee two	ay squared plus bee squared is equal to cee squared
$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}$	minus one list with one item ay bee cee dee	two lines line one ay bee line two cee dee to the negative one power
$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	three four one two zero one one one	the two by two matrix row one one two row two three four times the two by two matrix row one one one row two zero one